

Introduction to Linux (1) — Exercises

Lab 1 — Basic commands `ls`, `cd`, `mkdir`, `rm`, ...

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Part 1 of 2

The basics of the terminal

Commands useful for the following exercises:

```
cd  
cp  
hostname  
ls  
less  
mkdir  
mv  
pwd  
rm  
ssh  
uname  
wc
```

Commands are briefly explained in [linux_commands.pdf](#)

Exercise 0: Find out about commands

Find out *how to find out*:

Strategy: Look in `linux_commands.pdf`; the lecture notes (`linux_1.pfd`); search the net; just try something; ask somebody...

Last resort: There are solutions in `linux_1_solutions.pdf`

Also, try the following command:

`man <COMMAND>`



where you substitute `<COMMAND>` with the name of an actual command:

`man uname`

Scroll man-pages using the  and  keys or space.

Exit by  (*quit*)

Start the terminal

Open the terminal on the computer in front of you:

On a Mac, click the search icon (magnifying glass) in the upper right corner, and type `terminal`

On other computers, find out for yourself (e.g. on Windows, you might have to install Linux in some form)

The terminal offers a prompt (command line), where you type all commands

Exercise 1 System info

1. What command is used to get the name of the computer you are logged into? Try it out
 2. What command is used to get the name of the *operating system* (OS) of the computer? Try it
 3. Add the *-a switch* (flag, option) to the above command. What do you see?
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1. `hostname`
2. `uname`
3. `uname -a`
(You'll see details of the OS type, kernel version, etc)

Exercise 2 Log in to remote server

1. What is the command for logging into another computer (remote server)?
 2. How do you log into `triton.dsv.su.se` from the terminal?
 3. Log into `triton.dsv.su.se`
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1. `ssh`
2. `ssh <SUKAT-ID>@triton.dsv.su.se` where you replace `<SUKAT-ID>` with your own user id. Notice the `@` character
3. `ssh kali2818@triton.dsv.su.se` (replace `kali2818` with your id. You'll be prompted for your SU password)

Exercise 3 Repeat system info, different machine

After logging into `triton.dsv.su.se` (see above):

1. Run the command `hostname`. What do you see?
 2. Run the command `hostname -A` (uppercase A). What do you see?
 3. Run the command `uname -a` (lowercase a). What do you see?
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1. `hostname` 

`triton`

2. `hostname -A` 

`triton.dsv.su.se`

3. `uname -a` 

`Linux triton 6.1.0-23-amd64 ... Debian 6.1.99-1`
`... (edited)`

Exercise 4 The basics: list files

1. What is the command for listing the files in a directory? Run it
 2. List all files in the `/bin` directory
 3. List the contents of user `kali2818`'s home directory
 4. List the contents of `texts/` in `kali2818`'s home directory
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1. `ls` 
2. `ls /bin/` (or `ls /bin`)
3. `ls ~kali2818/` or `ls /home/dsv/kali2818/`
4. `ls ~kali2818/texts/` or `ls /home/dsv/kali2818/texts/`

Exercise 5 The basics: move about

1. What is the command for printing the path to the current directory? Run it
 2. What is the command for changing to another directory?
 3. Move (change) to the /tmp directory
 4. Run the command printing the current directory again (see 1.)
 5. How do you move *back* from /tmp to your *home* directory?
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1. pwd 
2. cd
3. cd /tmp
4. pwd (Notice the different output from 1.)
5. cd  (Followed by return, cd always brings you back home!)

Exercise 6 Create new directory

(Run `cd`  to ensure that you are home.)

1. What command creates a new directory?
 2. In your home directory, make a new directory called `lab1`
 3. List the contents of the current directory, to verify that `lab1` is now there (there should also be a directory called `public_html`)
 4. List the contents of the new `lab1` directory (it should be empty)
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1. `mkdir`
2. `mkdir lab1`
3. `ls`
4. `ls lab1/` (or `ls lab1`)

Exercise 7 Copy a file (1/2)

1. What command is used for copying files?
 2. Copy the file `ksl.txt` from the directory `/home/dsv/kali2818/texts/` (or shorter `~kali2818/texts/`) to your newly created `lab1` dir.
 3. Verify that the file `ksl.txt` is now in `lab1`
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1. `cp`
2. `cp /home/dsv/kali2818/texts/ksl.txt lab1/`
3. `ls lab1/`

Exercise 8 Copy a file (2/2)

1. Change directory to lab1/
2. Copy the file
 `~kali2818/texts/deeper_understanding.txt` to your
 current directory (lab1)

Hint: What character is used for “put it here” (to the current directory)?

1. `cd lab1/`
2. `cp ~kali2818/texts/deeper_understanding.txt .`

Notice the dot at the end of the command, “put the file right here, in the current directory”

Exercise 9 Rename a file

1. What command is used for moving and/or renaming files?
 2. Rename the file `ks1.txt` (in directory `lab1/`) to `poem.txt`
(If you happen to be in a different directory than `lab1/`: first do `cd ~/lab1/`)
 3. List files, to ensure that there is now a file `poem.txt`, but no `ks1.txt`
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1. `mv`
2. `mv ks1.txt poem.txt`
3. `ls`

Exercise 10 `cp` vs. `mv`

1. What is the difference between `cp` and `mv`?
 2. What is the difference between copying and moving a file?
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1. `cp` copies files, `mv` moves and/or renames files
2. `cp`: When you copy, the original file stays intact. You now have two different files with the same contents.
`mv`: After moving a file, the original file is no longer there

Exercise 11 Viewing the contents of a file

1. What commands can you use to view the contents of a text file?
 2. Take a look at the contents of `poem.txt` (previously `ks1.txt`, before you renamed it)
 3. What's in the file `deeper_understanding.txt`?
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1. `cat`, `more`, `less` or e.g. the micro text editor
2. `cat poem.txt` or
`more poem.txt` or
`less poem.txt` or even
`micro poem.txt` (a text editor; `ctrl` + `q` to quit)
3. The lyrics to a song by Kate Bush about computers, of course!

Exercise 12 Removing files

1. What command is used for deleting files?
 2. Return to your home directory
 3. Remove all files with the .txt extension in the directory lab1/
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1. `rm`
2. `cd`
3. `rm lab1/*.txt` (Or individually: `rm lab1/poem.txt`, etc)

Exercise 13 Copy a directory

1. Copy the directory `/home/dsv/kali2818/texts/` and its contents to `lab1/` (in your home directory)
 2. List the contents of `lab1/` to ensure that the `texts/` directory is there
 3. List the contents of the `texts/` directory
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1. `cp -r ~kali2818/texts/ lab1/` (The `-r` switch is crucial)
2. `ls lab1/`
3. `ls lab1/texts/`

Exercise 14 Count lines

1. Is the `wc` command short for “water closet”?
 2. Count the number of lines in the file `lab1/texts/ks1.txt`
 3. Count the total number of lines in the files with a `.txt` extension in `lab1/texts/`
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1. Don't be stupid.
2. `wc -l lab1/texts/ks1.txt`
3. `wc -l lab1/texts/*.txt`
or `cat lab1/texts/*.txt | wc -l`

Exercise 15 man to find help

1. The `wc` command has a switch (flag, option) that makes it print a help message. Use the command
`man wc`
to find out what that switch is. You might have to scroll down to find it.
Run `wc` with the switch.

(Many commands have this switch. Try to run another command with it.)

1. `wc --help` (Double hyphens)

Exercise 16 Scroll through output

Some commands generate more output than there may be room for in the terminal. One example is

```
python3 --help
```

which prints a long help message. (`python3` is used for running Python programs)

1. Run the command, to see what it looks like
2. Figure out a way to make it possible to scroll through the output, top to bottom

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1. `python3 --help` (You will only be able to see the tail of a long text)
 2. `python3 --help | less` or
 `python3 --help | more`

Exercise 17 `rm`: file vs directory

1. Use `mkdir` to create a new directory:
`mkdir testdir`
2. Why doesn't the following command work for deleting `testdir/`?
`rm testdir/`

To remove a directory and its contents, you should use the `-r` switch:

```
rm -r testdir/
```

(To remove a *file*, you don't need the `-r` switch.)

Exercise 18 Browsing a text file

1. What command can you use to browse a text file, e.g.
`/home/dsv/kali2818/texts/fk.txt`
 2. How do you scroll down the text?
 3. How do you exit?
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1. `less /home/dsv/kali2818/texts/fk.txt`
or
`more /home/dsv/kali2818/texts/fk.txt`
2.  or space
3. 

Exercise 19 . . : The Only Way Is Up¹

(You can run the `cd` command, without arguments, to ensure that you are in your home directory.)

Given that you are in your home directory, what does the following command do?

```
ls ../
```

It lists the contents of the directory above the one you are currently in.

If you are in your home directory on `triton.dsv.su.se`, this will be `/home/dsv/`.

In other words, you'll be listing the home directories of all users.

¹Yazz (1988)